MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, 90-DEGREE E-PLANE BEND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.

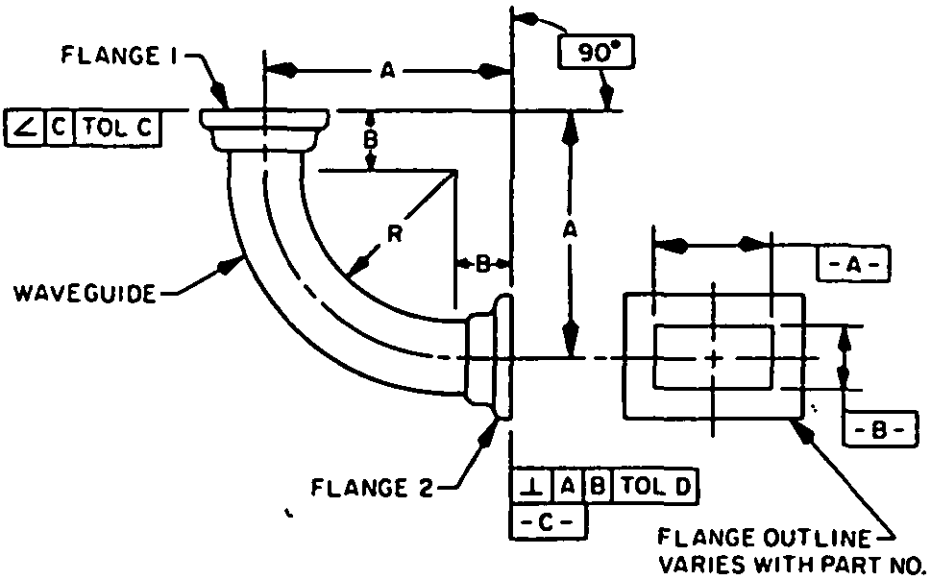


FIGURE 1. Assembly configuration.

FSC 5985

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TABLE I. Dash number, electrical and physical characteristics, and dimensions.

Dash No.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss	VSWR	Pressurization lb_f/in^2 gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
001	1-043	56-001	61-002	2.6 - 3.95	-	1.03	15	10.500 (266.70) ±.030 (.76)	2.250 (5.715)	.115 (2.92)	.012 (.30)	7.500 (190.50) ±.125 (3.18)
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	30	7.000 (177.80) ±.030 (.76)	1.500 (38.10)	.077 (1.96)	.008 (.20)	5.000 (127.00) ±.125 (3.18)
003	1-067	55-001	60-001	5.85 - 8.2	-	1.05	30	4.875 (123.83) ±.025 (.64)	1.125 (28.58)	.056 (1.42)	.006 (.15)	3.375 (85.72) ±.125 (3.18)
004	1-073	53-002	59-007	7.05 - 10	-	1.05	30	4.062 (103.17) ±.020 (.51)	1.000 (25.40)	.044 (1.12)	.005 (.13)	2.751 (69.88) ±.063 (1.60)
005	1-079	53-001	59-006	8.2 - 12.4	-	1.05	30	3.562 (90.47) ±.020 (.51)	0.812 (20.62)	.038 (.96)	.004 (.10)	2.500 (63.50) ±.063 (1.60)
006	1-102	54-001	59-003	18 - 26.5	.11	1.06	30	1.722 (43.74) ±.015 (.38)	0.535 (13.59)	.019 (.48)	.002 (.05)	1.063 (27.00) ±.063 (1.60)
007	1-106	54-001	59-003	18 - 26.5	.11	1.08	30	1.722 (43.74) ±.015 (.38)	0.535 (13.59)	.019 (.48)	.002 (.05)	1.063 (27.00) ±.063 (1.60)
008	1-077	53-003	59-008	8.2 - 12.4	-	1.05	30	3.562 (90.47) ±.020 (.51)	0.812 (20.62)	.038 (.96)	.004 (.10)	2.500 (63.50) ±.063 (1.60)
009	1-071	53-004	59-009	7.05 - 10	-	1.05	30	4.062 (103.17) ±.020 (.51)	1.000 (25.40)	.044 (1.12)	.005 (.13)	2.750 (69.85) ±.063 (1.60)
010	1-041	56-002	61-001	2.6 - 3.95	-	1.03	15	10.500 (266.70) ±.030 (.76)	2.250 (57.15)	.115 (2.92)	.012 (.30)	7.500 (190.50) ±.125 (3.18)

See footnotes at end of table.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued.

Dash No.	Waveguide MBS/	Flange 1	Flange 2	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lbf/in² gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
		M3922/		GHz	dB							
011	1-089	53-005	59-001	12.4 - 18	.07	1.05	30	2.601 (66.07) ±.015 (.38)	0.655 (16.64)	.128 (.71)	.003 (.08)	1.750 (44.45) ±.063 (1.60)
012	1-053	57-001	62-001	3.95 - 5.85	-	1.05	30	7.000 (177.80) ±.030 (.76)	1.500 (38.10)	.076 (1.93)	.008 (.20)	5.000 (127.00) ±.125 (3.18)
013	3-006	54-003	59-005	26.5 - 40	.15	1.08	30	1.242 (31.55) ±.010 (.25)	0.382 (9.70)	.017 (.43)	.001 (.03)	.749 (19.02) ±.063 (1.60)
014	1-065	55-002	60-002	5.85 - 8.2	-	1.05	30	4.875 (123.83) ±.025 (.64)	1.125 (28.58)	.056 (1.42)	.006 (.15)	3.375 (85.72) ±.125 (3.18)
015	1-093	53-005	59-001	12.4 - 18	.07	1.05	30	2.601 (66.07) ±.015 (.38)	0.655 (16.64)	.028 (.71)	.003 (.08)	1.750 (44.45) ±.063 (1.60)
016	1-103	54-002	59-004	18 - 26.5	.11	1.06	30	1.722 (43.74) ±.015 (.38)	0.535 (13.59)	.019 (.48)	.002 (.05)	1.063 (27.00) ±.063 (1.60)
017	3-005 3-008	54-003	54-003	26.5 - 40	.15	1.1	30	1.000 (25.40) ±.015 (.38)	.500 (12.70)	.013 (.33)	.001 (.03)	.500 (12.70) ±.015 (.38)
018	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	---	↓	↓	.014 (.36)	↓	↓
019	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	---	↓	↓	.014 (.36)	↓	↓
020	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	---	↓	↓	.014 (.36)	↓	↓
021	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	---	↓	↓	.014 (.36)	↓	↓
022	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	---	↓	↓	.012 (.30)	↓	↓

See footnotes at end of table.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued.

Dash No.	Waveguide MIL/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lbf/in2 gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
023	1-001 1-002 1-161	76-07	76-07	.32 - .49	.10	1.05	.25	27.000 (685.80) ±.015 (.38)	3.000 (76.20)	.302 (7.67)	.102 (2.59)	18.062 (458.77) ±.015 (.38)
024	1-003 1-004 1-162	76-06	76-06	.35 - .53	.10	1.05	.25	27.000 (685.80) ±.015 (.38)		.297 (7.54)	.093 (2.36)	18.562 (471.47) ±.015 (.38)
025	1-005 1-006 1-163	76-05	76-05	.410 - .625	.10	1.05	.25	21.000 (533.40) ±.015 (.38)		.238 (6.05)	.080 (2.03)	13.375 (339.72) ±.015 (.38)
026	1-007 1-008 1-164	76-04	76-04	.490 - .750	.10	1.05	.25	15.000 (381.00) ±.015 (.38)		.179 (4.55)	.067 (1.70)	8.125 (206.38) ±.015 (.38)
027	1-009 1-010 1-165	76-03	76-03	.64 - .96	.10	1.05	.50	15.000 (381.00) ±.015 (.38)		.171 (4.34)	.051 (1.30)	9.000 (228.60) ±.015 (.38)
028	1-011 1-012 1-166	76-02	76-02	.75 - 1.12	.10	1.05	.50	12.000 (304.80) ±.015 (.38)		.141 (3.58)	.044 (1.12)	6.437 (163.50) ±.015 (.38)
029	1-013 1-014 1-167	76-01	76-01	.96 - 1.45	.10	1.05	1.0	8.000 (203.20) ±.015 (.38)	2.000 (50.80)	.102 (2.59)	.035 (.89)	3.950 (100.33) ±.015 (.38)
030	1-018 1-019 1-168	52-002	52-002	1.12 - 1.70	.10	1.05	1.5	8.000 (203.20) ±.015 (.38)		.092 (2.34)	.029 (.74)	4.295 (109.09) ±.015 (.38)
031	1-029 1-030 1-170	52-006	52-006	1.70 - 2.60	.10	1.05	3.0	8.000 (203.20) ±.015 (.38)		.079 (2.01)	.019 (.48)	4.845 (123.06) ±.015 (.38)
032	1-075 1-079	53-001	53-001	8.20 - 12.4	.10	1.05	30	1.438 (36.53) ±.015 (.38)	.688 (17.48)	.027 (.69)	.004 (.10)	.500 (12.70) ±.015 (.38)

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Electrical characteristics: See table I.

Part number: M3970/7-(dash number from table I).

Cross reference: See table II.

TABLE II. Cross reference.

Part number M3970/7-	Type designation	Part number M3970/7-	Type designation
001	UG-860/U	009	UG-868/U
002	UG-861/U	010	UG-869/U
003	UG-862/U	011	UG-870/U
004	UG-863/U	012	UG-871/U
005	UG-864/U	013	UG-872/U
006	UG-865/U	014	UG-873/U
007	UG-866/U	015	UG-874/U
008	UG-867/U	016	UG-875/U

NOTE: Assemblies covered by this specification are substitutable for the following commercial types:

Part number M3970/7-	Manufacturer's code	Commercial type
017	73293	44321H-901
018	73293	44322H-901
019	73293	44323H-901
020	73293	44324H-901
021	73293	44325H-901
022	73293	44326H-901
023	52564	3004001
024	52564	3104001
025	52564	3204001
026	52564	3304001
027	52564	3404001
028	52564	3504001
029	52564	3604001
030	52564	3704001
031	52564	4004001
032	06424	1202012-1
032	83099	TR642427
032	00929	X623B

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Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DSA - ES

User activities:
Army - MU
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0835-4)

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